Assessed Project — Part 1: Understanding the Small-Data Problem

Weighting: 30% of the assessed project

Aim

This part of the assessed project is designed to help you understand the challenges of machine learning with small datasets and to develop a rigorous experimental framework that you will build on in Part 2.

Research question

How does limited training data affect the performance, stability, and reliability of machine-learning classifiers?

Task

Using the datasets provided, investigate how a selection of standard classifiers behaves as the amount of available training data is progressively reduced.

Your study should consider:

- predictive performance as sample size decreases;
- stability and variability across repeated runs or data splits;
- sensitivity of different classifiers to small sample sizes;
- evidence of overfitting or unreliable model comparison;
- the effect of experimental choices such as validation strategy and model complexity.

You should compare several appropriate baseline classifiers and use a fair and consistent evaluation protocol.

You are expected not simply to report classifier accuracies, but to use your experiments to understand what changes when training data become scarce, why these changes occur, and how confidently conclusions can be drawn from small datasets.

What do we mean by “small”?

There is no single sample size that defines a dataset as “small”. Whether a training set is small depends on factors such as the number of features, number of classes, class balance, complexity of the underlying problem, and the learning algorithm being used.

You should therefore investigate a range of training-set sizes and justify your choices.

As a general guide:

- use several progressively smaller training-set sizes for each dataset;
- include sizes that span from a reasonably well-sampled setting to a clearly data-scarce setting;
- ensure that all classes remain adequately represented wherever possible;
- use repeated, preferably stratified, sampling or cross-validation to investigate variability;
- report not only average performance but also an appropriate measure or visualisation of variability.

The exact sample sizes do not need to be identical across datasets. Choosing and justifying an appropriate experimental design is part of the task.

Learning curves showing performance and variability against training-set size are strongly encouraged.

Datasets

The following datasets are provided in Canvas:

- Iris
- Wine
- Breast Cancer Wisconsin (Diagnostic)
- Optical Digits (scikit-learn subset)
- Glass2
- Chess KR-vs-KP
- Zoo

These datasets deliberately differ in sample size, dimensionality, number of classes and feature characteristics.

You should use the supplied datasets for the main study. They should form the basis of the experimental framework that can subsequently be reused in Part 2.

You are encouraged to examine results across all datasets, but you do not need to discuss every individual experiment in equal depth. You may select particularly informative datasets for more detailed analysis and visualisation, provided that the broader comparative evidence is still presented.

Any preprocessing, feature scaling, encoding, feature selection or parameter tuning must be incorporated correctly into the evaluation procedure. In particular, information from the test data must not be used when fitting preprocessing steps or selecting models.

Expected outcome

Your submission should provide evidence-based answers to questions such as:

- How does classifier performance change as data become scarce?
- Which types of models appear more or less robust to small sample sizes?
- What evidence is there of overfitting or unreliable model selection?

- How reliable are conclusions drawn from very small datasets?
- What constitutes a sound experimental protocol for small-data analysis?

There is not necessarily one universally “best” classifier. The emphasis is on understanding and explaining the behaviour you observe.

Submission

Submit a Jupyter notebook (`.ipynb`) containing your experimental code, results, figures and Markdown commentary. Your notebook should be clear and reproducible and should normally contain:

- a short introduction and research questions;
- description and justification of the experimental protocol;
- classifiers and datasets used;
- results, including appropriate tables and visualisations;
- analysis and discussion of the findings;
- conclusions about learning from small datasets.

As a guideline, aim for approximately 1500-2000 words of explanatory Markdown commentary. The word count should exclude code, figures and tables, figure/table captions, and references. The emphasis is on the quality of the experimental evidence and analysis, rather than reaching a particular word count.

A separate video presentation is not required for Part 1.

Link to Part 2

Part 2 of the assessed project will build directly on this work.

You will be asked to investigate a probability-based machine-learning approach intended to provide an alternative way of learning from limited data, evaluate whether it addresses some of the problems identified in Part 1, and explore how that approach can be optimised to achieve the best results.

The datasets and experimental framework developed here should therefore be designed so that they can be reused and extended in Part 2.

Your Part 1 work should establish:

- baseline results;
- a sound and reproducible evaluation protocol;
- appropriate measures of performance and stability; and
- a clear understanding of the problems caused by limited training data.

You are not expected to investigate the Part 2 method in Part 1.

Suggested starting references

The following recent papers provide useful starting points for understanding machine learning with limited data, learning curves, and reliable model evaluation:

- Naser, M. Z. (2026). *A review of machine learning with small and limited data*. Journal of Big Data, 13, 18.
- Steinert, S. et al. (2024). *A refined approach for evaluating small datasets via binary classification using machine learning*. PLOS ONE, 19(5), e0301276.
- Silvey, S. & Liu, J. (2024). *Sample Size Requirements for Popular Classification Algorithms in Tabular Clinical Data: Empirical Study*. Journal of Medical Internet Research, 26, e60231.
- Viering, T. & Loog, M. (2023). *The Shape of Learning Curves: A Review*. IEEE Transactions on Pattern Analysis and Machine Intelligence, 45(6), 7799–7819.

Students are encouraged to identify and cite additional relevant literature.

Key principle

Part 1 asks you to understand the problem. Part 2 will ask you to investigate and optimise a possible solution.

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Assessment criteria and marking

Part 1 will be marked out of 100 and contributes 30% of the assessed project. The mark is based on the quality of the investigation, not on achieving a particular accuracy or identifying a particular "best" classifier.

Criterion	Weight	What is assessed
Experimental design and methodological rigour	25%	Choice and justification of training-set sizes; repeated and, where appropriate, stratified evaluation; consistent comparison of classifiers; correct handling of preprocessing, tuning and validation; avoidance of information leakage.
Results and presentation of evidence	20%	Sufficient evidence across datasets and sample sizes; appropriate performance measures; reporting of variability; clear tables, figures and learning curves.
Analysis and interpretation	30%	Explanation of how classifier performance and stability change as data become scarce; comparison of robustness and overfitting; interpretation of variability and limits of the evidence.
Understanding and conclusions	15%	Synthesis across experiments; evidence-based answer to the research question; recognition of limitations; identification of a sound small-data evaluation protocol and implications for Part 2.
Reproducibility, communication and notebook quality	10%	Runnable and clearly organised notebook; readable code and Markdown; well-labelled figures and tables; appropriate references.

Marking rubric

The criteria above are applied using the overall standards below. The descriptors indicate the typical quality of work in each band; the final mark reflects performance across all criteria.

Mark band	Typical standard
70–100 Distinction	Strong, well-justified design; substantial and clearly presented evidence, including variability; analytical discussion that explains observed patterns and limitations; conclusions closely supported by the results; clear and reproducible notebook. Marks of 80+ require exceptional depth, rigour and synthesis; novelty is not required.
60–69 Merit	Sound design and good evidence; clear interpretation of the main patterns; conclusions supported by the results. Some limitations in depth, justification, treatment of variability or synthesis.
50–59 Pass	Adequate investigation addressing the research question; generally workable design and sufficient evidence. Analysis is mainly descriptive or uneven, and treatment of variability, limitations or justification is incomplete.
40–49 Marginal fail	Investigation is attempted, but significant weaknesses in design, evidence or interpretation limit confidence in the conclusions. Important issues such as repeated evaluation, consistency of comparison, variability or leakage control may be inadequately handled.
Below 40 Fail	Work is substantially incomplete or methodologically unsound, with insufficient reliable evidence or analysis to address the research question.